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EXAMPLE 9.4

Calculate the rms speed, average speed and the most probable speed of 1 mole of hydrogen molecules at 300 K. Neglect the mass of electron.

Solution

The hydrogen atom has one proton and one electron. The mass of electron is negligible compared to the mass of proton.

Mass of one proton = $1.67 \times 10^{-27} kg$.

One hydrogen molecule = 2 hydrogen atoms = $2 \times 1.67 \times 10^{-27} kg$.

The average speed

$$\bar{v} = \sqrt{\frac{8kT}{\pi m}} = 1.60\sqrt{\frac{kT}{m}} =$$

$$= 1.60\sqrt{\frac{(1.38\times10^{-23})\times(300)}{2(1.67\times10^{-27})}} = 1.78\times10^{3}\, m s^{-1}$$

(Boltzmann Constant k = $1.38 \times 10^{-23}\ J\ K^{-1}$)

The rms speed $v_{rms} = \sqrt{\frac{3kT}{m}} = 1.73\sqrt{\frac{kT}{m}}$

$$= 1.73\sqrt{\frac{(1.38\times10^{-23})\times(300)}{2(1.67\times10^{-27})}} = 1.93\times10^{3}\, m s^{-1}$$

Most probable speed $v_{mp} = \sqrt{\frac{2kT}{m}} = 1.41\sqrt{\frac{kT}{m}}$

$$= 1.41\sqrt{\frac{(1.38\times10^{-23})\times(300)}{2(1.67\times10^{-27})}} = 1.57\times10^{3}\, m s^{-1}$$

Note that $v_{rms} > \bar{v} > v_{mp}$

9.2.8 Maxwell-Boltzmann speed distribution function

In a classroom, the air molecules are moving in random directions. The speed of each molecule is not the same even though macroscopic parameters like temperature and pressure are fixed. Each molecule collides with every other molecule and they exchange their speed. In the previous section we calculated the rms speed of each molecule and not the speed of each molecule which is rather difficult. In this scenario we can find the number of gas molecules that move with the speed of 5 m s^{-1} to 10 m s^{-1} or 10 m $s^{-1}$to 15 m s^{-1} etc. In general our interest is to find how many gas molecules have the range of speed from v to $v + dv$. This is given by Maxwell's speed distribution function.

$$N_v = 4\pi N\left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2kT}} \qquad (9.24)$$

The above expression is graphically shown as follows

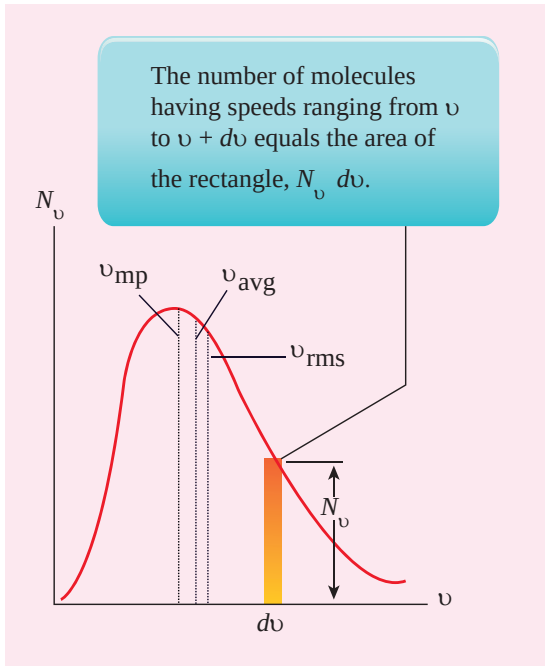


Figure 9.3 Maxwell's molecular speed distribution

From the Figure 9.3, it is clear that, for a given temperature the number of molecules having lower speed increases parabolically but decreases exponentially after reaching most probable speed. The rms speed, average speed and most probable speed are indicated in the Figure 9.3. It can be seen that the rms speed is greatest among the three.